# From Comprehension to Compliance: How Post-Training Couples Moral Representations to Model Behavior in Language Models

**Orion Reblitz-Richardson***

## Abstract

Does a language model that behaves morally understand morality, or has it only learned to comply? We separate moral *comprehension* (what a model internally represents) from *compliance* (how it behaves) and track both across the full OLMo-3 7B alignment pipeline, 25 model states spanning base, 13 pre-training checkpoints, SFT, DPO, and RLVR, plus a refusal-ablated model. Moral comprehension is a pre-training property: the six Moral Foundations are linearly decodable at 100%, span five effective dimensions, and transfer as near-perfect classifiers at every stage, with the moral subspace crystallizing during pre-training (direction cosine to the final base $0.87 \rightarrow 0.999$). Post-training reorients rather than re-teaches: SFT rotates the subspace $\sim 40^\circ$ once, while DPO and RLVR leave it unchanged. Comprehension and behavioral compliance are only weakly coupled ($\phi \approx 0.05$ even in the fully aligned model), which predicts that compliance is a separable mechanism. A Heretic-style refusal direction is nearly orthogonal to the moral subspace (projection fraction $0.10$); ablating it removes refusal while leaving comprehension and moral judgment intact. We then test whether this separation can be repaired: a training-time ablation-resistance loss deepens the model's reliance on its moral subspace but cannot move the compliance mechanism into it, because the refusal direction forms during post-training from pre-training features already orthogonal to moral structure. Alignment is a wiring step, not a teaching step: it couples a separable compliance mechanism to moral representations that pre-training already built, and making that wiring robust to ablation is a pre-training problem, not a post-training one. A direct pre-training intervention bears this out only halfway: a regularizer places the refusal contrast inside the moral subspace geometrically, but the coupling does not survive fine-tuning as a functional dependence.

## 1 Introduction

Does a language model that behaves morally understand morality, or has it only learned to comply? Earlier papers in this series studied what models *represent*: moral content is linearly decodable early in pre-training and across nearly all layers (Reblitz-Richardson, 2026c), is encoded uniformly across mixture-of-experts experts rather than specialized (Reblitz-Richardson, 2026b), and organizes into structured, five-dimensional framework geometry whose directions are causal for moral judgments (Reblitz-Richardson, 2026a,d). All of that was measured on base, pre-trained models. The moral behavior we actually deploy and align, refusing harmful requests and giving morally-aware answers, is produced by *post-training*: supervised fine-tuning (SFT), preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR). This paper asks what post-training does to moral representation. Does alignment *teach* morality, or does it *wire* a pre-existing moral representation to behavior?

*Distiller Labs. Correspondence to Distiller Labs <orion@orionr.com>.

Preprint.

We separate two things a model can have. **Comprehension** is the moral content a model internally represents: which foundation a scenario engages, how foundations relate, whether the representation is decodable and stable. **Compliance** is whether the model's behavior conforms: whether it refuses a harmful request, whether its stated judgment matches the morally expected one. A model can have either without the other. Our hypothesis is that alignment is a **coupling change, not a teaching change**: post-training does not create moral understanding, it connects understanding that pre-training already built to behavioral output. If so, failures of alignment, jailbreaks and ablations, are failures of *coupling*, not of *understanding*.

We test this on the full OLMo-3 7B pipeline, which publishes every stage: the base model and 13 stage-3 pre-training checkpoints, the SFT and DPO snapshots, and the final RLVR-tuned Instruct model with 8 intermediate checkpoints, 25 model states in all, plus a refusal-ablated instruct. Across them we measure comprehension (foundation probe-direction transfer and geometry), compliance (behavioral moral judgment and refusal), the coupling between them, the persona direction and its angle to the moral subspace, and the geometry of the refusal direction itself. The results form a chain, in which each finding raises the question the next one answers.

**Finding 1: Moral comprehension is a pre-training property.** Across all 25 states the six foundations are linearly decodable at 100%, span five effective dimensions, and base-trained directions transfer as near-perfect classifiers (AUC $\approx 1.0$). The moral subspace *crystallizes* during pre-training: the cosine between a checkpoint's foundation directions and the final base model's rises monotonically from $0.87$ to $0.999$ over the stage-3 anneal. Comprehension is in place before any alignment.

**Finding 2: Post-training reorients moral representation, it does not re-teach it, and SFT does so once.** SFT applies a single $\sim 40°$ rotation to the moral subspace (direction cosine to base $0.999 \rightarrow 0.757$); DPO and all eight RLVR checkpoints leave it flat. Effective dimensionality stays at five, mean pairwise foundation cosine barely moves ($0.262 \rightarrow 0.250$), and the framework clustering is preserved. Alignment rotates the moral representation slightly and then leaves it alone.

**Finding 3: Comprehension and compliance are only weakly coupled, and that weak coupling is itself the result.** Per-scenario agreement between the model's internally dominant foundation and its behavioral correctness rises across post-training ($0.375 \rightarrow 0.500$; $\phi$ $-0.19 \rightarrow +0.05$) but remains weak: even at the final Instruct model, $P(\text{comply} \mid \text{comprehend}) \approx P(\text{comply} \mid \neg\text{comprehend}) \approx 0.75$. A near-zero coupling is not a null result; it states that in the fully aligned model, moral representation and behavioral compliance are barely linked. This predicts a specific, testable consequence: if compliance were routed *through* moral representations, removing compliance would damage comprehension; weak coupling predicts instead that compliance is a *separate* mechanism that can be removed while comprehension survives.

**Finding 4: The refusal mechanism is geometrically separate from morality, and removing it confirms the dissociation.** The refusal direction recovered by Heretic-style difference-of-means (Arditi et al., 2024) is nearly orthogonal to the six-foundation moral subspace (projection fraction $0.10$, mean $|\cos| = 0.06$). Ablating it leaves comprehension untouched (effective dimensionality 5, direction transfer and probe accuracy identical to the un-ablated instruct) and moral judgment intact (behavioral accuracy $0.75 \rightarrow 0.73$), while refusal collapses (refusal rate $0.25 \rightarrow 0.00$): the ablated model now answers requests it previously refused. This is the high-comprehension, low-compliance cell of the dissociation, realized in a real model.

**The separation is a vulnerability, and it cannot be repaired after pre-training.** If compliance is a separable mechanism, the obvious defense is to couple it back to comprehension, so that the refusal-direction ablation of Finding 4 can no longer remove one without the other. We first measure how much the model relies on its moral subspace for generation, a reliance that is present at the base and grows roughly sixfold across alignment, then attempt to deepen it with ablation-resistance training: an auxiliary fine-tuning loss that rewards the model when projecting the moral subspace out of its activations degrades its output. The intervention builds strong dependence on the moral subspace but does not couple compliance to it. That dependence is orthogonal to the refusal direction an attacker removes, and refusal ablation damages the trained model no more than the control. The failure is developmental: the refusal feature forms during post-training out of pre-training proto-features that are already orthogonal to the moral subspace, so a post-training loss acts too late to relocate it. Making alignment ablation-resistant is, on this evidence, a pre-training problem. We then run that pre-training intervention directly: a continued-pre-training regularizer places the refusal contrast geometrically inside the moral subspace, but the coupling does not survive fine-tuning as a functional dependence, and the separation holds. Geometric coupling is installable; functional ablation-resistance is not.

Together these findings populate a comprehension×compliance matrix in which the low-comprehension row is empty, every OLMo-3 state has full moral decodability, and all variation lives in the high-comprehension row, where compliance moves independently of understanding. The contribution is twofold: a method for measuring comprehension, compliance, and their coupling across an alignment pipeline, with the result that alignment is a wiring step that preserves moral comprehension built in pre-training and attaches a separable compliance mechanism, so that compliance can be removed without touching what the model understands; and the finding that this separation cannot be undone after the fact, which locates the problem of ablation-resistant alignment in pre-training rather than post-training, where a direct intervention installs the coupling in the representation but does not make it functional.

## 2 Related Work

**Refusal directions and abliteration.** Arditi et al. (2024) showed that refusal in aligned language models is mediated by a single linear direction in the residual stream, recoverable as a difference-of-means between harmful and harmless prompts, and that orthogonalizing the weight matrices against it removes refusal. Heretic (p-e-w, 2025) packages this into an automatic decensoring tool with a fixed harmful/harmless prompt set and per-layer optimization. We use Heretic's prompt set and the difference-of-means direction, but ablate with Arditi et al.'s uniform single-direction method for a controlled experiment, and our question is not how well refusal can be removed but where the refusal direction sits relative to moral representations.

**Moral representation in language models.** A growing body of work probes whether and how models encode moral and ethical content (Hendrycks et al., 2021), typically as a binary moral/neutral distinction. This series extended that to structured representations: moral content is decodable early and broadly (Reblitz-Richardson, 2026c), diffuse across MoE experts (Reblitz-Richardson, 2026b), organized into five-dimensional framework geometry (Reblitz-Richardson, 2026d) whose directions are causal for judgments (Reblitz-Richardson, 2026a), all on base models. We carry that representational lens through post-training and to behavior. We use Moral Foundations Theory (Graham et al., 2013, Haidt, 2012) as the foundation inventory.

**Alignment depth.** Whether alignment is shallow has been debated since RLHF (Ouyang et al., 2022) and preference optimization (Rafailov et al., 2023) became standard. Work on alignment faking (Greenblatt et al., 2024) and the ease of jailbreaks and fine-tuning attacks suggests aligned behavior is brittle. Our contribution is mechanistic: we show *why* it is removable, by measuring the coupling between moral representation and behavior and locating the refusal mechanism outside the moral subspace.

**Persona features.** Wang et al. (2025) identified a toxic-persona latent that controls emergent misalignment. We recover a linear analog and track its relationship to moral representations across the pipeline, finding the persona direction stays nearly orthogonal to the moral subspace.

**Linear probing.** Our directions are linear probes (Alain and Bengio, 2017); the geometry and transfer methodology follows Reblitz-Richardson (2026d).

## 3 Methodology

### 3.1 Models and the pipeline grid

We study the OLMo-3 7B family (Ai2, 2025), which publishes every stage of its alignment pipeline. Our grid has 25 model states: the base model (`Olmo-3-1025-7B`) and 13 stage-3 pre-training anneal checkpoints (steps 1000–11921), the SFT and DPO snapshots, the final RLVR-tuned Instruct model, and its 8 intermediate RLVR checkpoints. To these we add one refusal-ablated Instruct model (Section 3.6). OLMo-3 7B is a 32-layer, 4096-dimensional decoder with hybrid attention: 24 sliding-window layers and 8 full-attention layers (layers 3, 7, . . . , 31). For the short probing texts used here every token attends within the window, so the attention pattern is inert for probing; we nonetheless flag the full-attention layers in all layer-wise plots and find no periodicity.

We probe foundation directions on the base model and transfer them across the grid. All probing uses raw-text inputs: a Sprint-1 comparison found that base-trained directions transfer to the instruct model with higher and more uniform agreement under raw text than under the chat template (mean direction cosine $0.94$ vs. $0.91$), and raw text is the only format defined consistently across the base and pre-training checkpoints, which have no chat template. Behavioral and coupling measurements, which require the model to respond, use the chat template.

### 3.2 Foundation probe directions and transfer

Following the geometry method of this series (Reblitz-Richardson, 2026d) and standard linear probing (Alain and Bengio, 2017), we train one linear probe per Moral Foundations Theory (Graham et al., 2013, Haidt, 2012) foundation on mean-pooled residual activations of matched moral/neutral sentence pairs, at each layer, with a fixed seed so the learned weight vector (the foundation "direction") is a deterministic function of the activations. We also record the mean-difference direction. We report two transfer quantities for a state: the ROC-AUC of the base direction applied to that state's activations (does the base direction still separate moral from neutral?), and the cosine between the base direction and a direction freshly fitted on that state (how much has the direction rotated?). Bootstrap resampling on the base model (200 resamples) gives a stable layer band of 15–31 across all foundations (Appendix B); geometry is reported over that band.

### 3.3 Framework geometry

From the six foundation directions at a layer we compute the $6 \times 6$ cosine matrix, the mean pairwise cosine (collapse if near 1, isolation if near 0), the effective dimensionality (principal components for $90\%$ variance), and a Ward clustering of $1 - \cos$. Tracking these across the grid measures whether post-training preserves or restructures the moral representation.

### 3.4 Coupling: comprehension vs. compliance

Comprehension and compliance can each be measured on their own; the question this paper turns on is how tightly the first predicts the second. We measure that link per scenario, on the 48 moral scenarios of the behavioral benchmark (six foundations $\times$ four difficulty levels: obvious, moderate, subtle, adversarial), each annotated with a target foundation and an expected approve/disapprove judgment, presented through the chat template.

**Comprehension signal (internal).** For a scenario we take the residual-stream activation at the stable layer (layer 16) at the final prompt token, the position from which the model begins its answer, and project it onto the six base-trained foundation directions. The six projections sit on different scales, so we $z$-score each foundation's projection across the 48 scenarios and define the dominant foundation as the $\arg\max$ of the $z$-scored projections. The scenario is *comprehended* if its dominant foundation matches the annotated target. This is a stricter test than probe accuracy: probe accuracy asks whether a foundation is decodable in aggregate, whereas this asks whether the model places *this* scenario on the right foundation at the decision point. Chance is $1/6 \approx 0.17$. We use the base-trained directions at every state, so comprehension is read on the same six axes throughout.

**Compliance signal (behavioral).** We greedily generate a completion and parse its moral judgment; the scenario *complies* if the judgment matches the expected one. We use this approve/disapprove judgment, not the comply/refuse classifier, because the former is parsed reliably from a direct "is this acceptable?' ' prompt while the latter is noisier on borderline requests (Appendix B).

**Coupling statistics.** Each scenario yields two bits, comprehended and complied, and a $2 \times 2$ contingency table over the 48 scenarios. We summarize it three ways: raw agreement (the fraction of scenarios on which the two bits coincide); the $\phi$ coefficient (the Pearson correlation between the two binary variables, equivalently the Matthews correlation of the table), for which $\phi = 0$ is statistical independence; and the conditional compliance rates $P(\text{comply} \mid \text{comprehend})$ and $P(\text{comply} \mid \neg\text{comprehend})$, whose difference is the predictive-gap form of the same relationship. Reading all three keeps the agreement number from being mistaken for coupling when it is driven by the base rates of the two bits.

**What the measure is and is not.** Coupling here is correlational: it asks whether the foundation the model internally activates co-varies, across scenarios, with whether it behaves correctly. It does not show that the moral representation *causes* the behavior; a causal test would steer the foundation direction during generation and read the change in judgment, which we leave to future work. With 48 scenarios and keyword judgment parsing the estimate is noisy, and difficulty level is a partial confound we do not control, so we report coupling qualitatively, as the result that the two are barely linked, rather than as a precise coefficient. Its value is comparative: applying the same procedure at SFT, DPO, Instruct, and the ablated model makes the trend in coupling and its small magnitude interpretable even when any single coefficient is imprecise.

### 3.5 Persona direction

We train a linear persona probe on persona-voice vs. neutral-voice minimal pairs (a linear analog of the toxic-persona feature of Wang et al., 2025) and, per layer, measure its accuracy and the cosine of the persona direction to each foundation direction (the persona–morality angle). Tracking both across the grid asks whether alignment couples persona to morality.

### 3.6 Refusal ablation and refusal–morality geometry

We recover a refusal direction with the Heretic protocol (Arditi et al., 2024, p-e-w, 2025): its exact prompt set (`mlabonne/harmful_behaviors` and `mlabonne/harmless_alpaca`, 400 train prompts each), the chat template with a generation prompt, and a per-layer difference-of-means of the final-token residual between harmful and harmless prompts. We ablate with Arditi et al.'s uniform single-direction method rather than Heretic's per-layer optimization, a controlled rather than maximal ablation: the unit refusal direction $\hat{d}$ from the stable layer is orthogonalized out of every layer's attention out-projection and MLP down-projection, $W \leftarrow W - \hat{d}\hat{d}^{\top}W$. We then re-run the full battery (Sections 3.2–3.5) on the ablated model. The refusal–morality geometry is the cosine of $\hat{d}$ to each foundation and the fraction of $\hat{d}$'s norm lying in the six-foundation span (a least-squares projection): a low fraction means refusal is carried outside the moral subspace.

Behavioral compliance uses two benchmarks: a moral-scenario judgment benchmark (per-foundation approve/disapprove accuracy) and a borderline-request refusal benchmark (the fraction of harmful/borderline requests the model declines). The refusal classifier flags an opening refusal as a refusal regardless of length; this corrects a failure mode in which a long "I'm sorry, but I can't . . . " decline was scored as compliance (Appendix B).

### 3.7 Moral-subspace dependency and ablation-resistance training

Two further measures (Section 6) ask whether moral comprehension is *used*, and whether its use can be strengthened into a coupling.

**Dependency.** Decodability says morality is present; dependency asks how much the model relies on it to generate. We project the six foundation directions out of the residual stream at every layer, the activation analog of the weight ablation in Section 3.6 (at each layer the hidden state is orthogonalized against an orthonormal basis of the foundation span), and measure the resulting increase in next-token cross-entropy on morally-loaded text relative to matched-neutral text:

$$\text{dependency} = \big(\text{CE}^{\text{abl}}_{\text{moral}} - \text{CE}_{\text{moral}}\big) - \big(\text{CE}^{\text{abl}}_{\text{neutral}} - \text{CE}_{\text{neutral}}\big).$$

The neutral arm controls for the generic cost of removing any six directions; the difference isolates the moral-specific reliance. We report it across the grid using the base foundation directions, comparable across states, and as a robustness check using each state's own directions.

**Ablation-resistance training (ART).** To test whether that dependency can be deepened into a coupling, we add an auxiliary loss during supervised fine-tuning. Each step runs a paired forward pass: the standard loss $L_{\text{sft}}$ and an ablated loss $L_{\text{abl}}$ computed with the foundation subspace projected out of every layer. The projection is differentiable in the activations (the frozen directions carry no gradient), so the gradient can teach the model to route generation through the moral subspace. A bounded hinge drives the gap up to a target $g^{\star}$ and then stops:

$$\text{ART} = \lambda \ \text{relu}\big(g^{\star} - (L_{\text{abl}} - L_{\text{sft}})\big).$$

The bound is necessary: the unbounded $-\lambda\,(L_{\text{abl}} - L_{\text{sft}})$ objective diverges, inflating $L_{\text{abl}}$ without limit and destroying the standard loss. The gap is measured on a pool of concentrated moral text, not

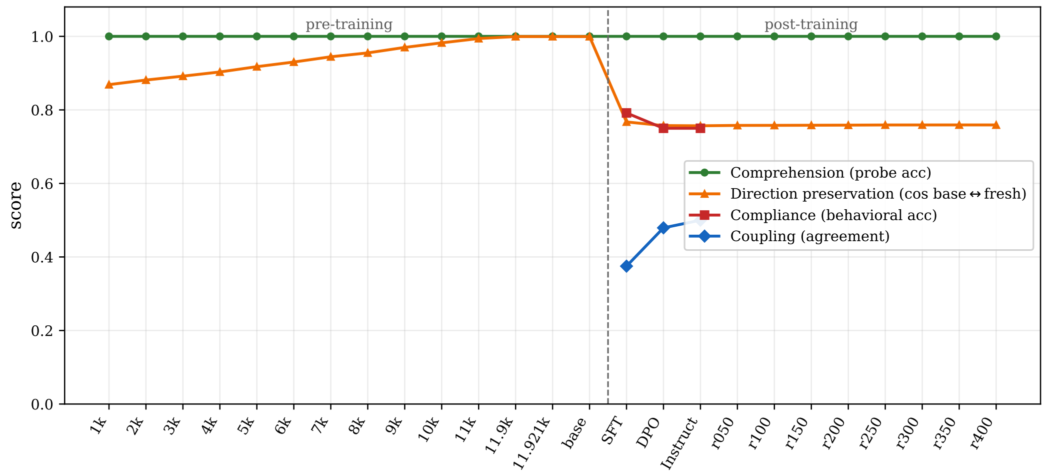


**Figure 1:** Comprehension, direction preservation, compliance, and coupling across the OLMo-3 pipeline (25 states, left to right: 13 stage-3 pre-training checkpoints, base, SFT, DPO, Instruct, 8 RLVR substeps). Comprehension (probe accuracy) is flat at 1.0 throughout. Direction preservation, the cosine of each state's foundation directions to the final base model, crystallizes during pre-training and drops once at SFT. Compliance and coupling are defined only for the instruct-capable states.

the mixed fine-tuning batch where general content dilutes it to a no-op (Section 6.3); $\lambda$ is calibrated so the ART term is $\sim 10\%$ of $L_{\text{sft}}$ on the first batch and capped at 1.

**ART-SFT and evaluation.** We fine-tune OLMo-3 base into an instruct model with LoRA (rank 16, query/value projections) on a general-plus-moral chat mixture (a Tülu-3 SFT subset with a moral-content supplement), in two conditions identical except for the ART term: a control ($\lambda = 0$) and ART. We then run the full battery (Sections 3.2–3.5, plus dependency) on each model and apply the Section 3.6 refusal ablation to both, yielding a four-cell comparison (control and ART, each with and without refusal ablation).

# 4 Results

We report the four findings as the chain laid out in the introduction. All probing uses raw-text inputs (Section 3); behavioral and coupling measurements use the chat template. Geometry metrics are reported over the bootstrap-stable layer band (layers 15–31; Section 3, Appendix B).

## 4.1 Moral comprehension is a pre-training property

At every one of the 25 model states, the six Moral Foundations are linearly decodable at $100\%$ (mean held-out probe accuracy $1.00$ across foundations), the foundation directions span five effective dimensions ($90\%$ variance), and the base-trained directions transfer to each state as near-perfect classifiers (mean $|\text{AUC}| \approx 1.0$). Comprehension does not emerge during post-training; it is fully present at the base and, as the pre-training trajectory shows, well before.

What *does* move during pre-training is the orientation of the moral subspace. The cosine between a checkpoint's foundation directions and the final base model's rises monotonically across the stage-3 anneal, from $0.869$ at step 1000 to $0.999$ by step 11921 (Figure 1, orange). The moral subspace crystallizes into its final form during pre-training: by the end of the anneal the foundation directions are essentially identical to the base model's, and their decodability is already saturated.

## 4.2 Post-training reorients moral representation, it does not re-teach it

If comprehension is already in place at the base, post-training cannot be teaching it. We ask instead how much post-training *moves* the representation. The answer is: once, at SFT, and only modestly. The cosine between the base foundation directions and each post-training state's freshly fitted directions drops from $0.999$ (base) to $0.757$ at SFT, a rotation of roughly $40°$. DPO ($0.757$) and

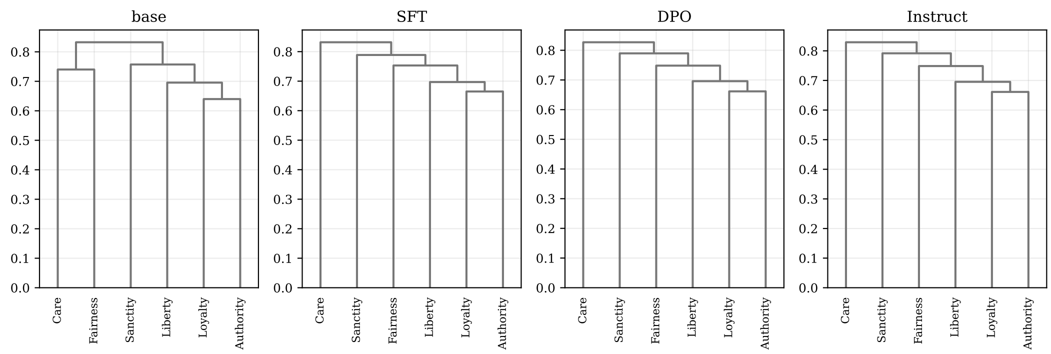


**Figure 2:** Foundation clustering (Ward linkage on $1 - \cos$) at base, SFT, DPO, and Instruct. The Loyalty–Authority binding pair persists across all stages. SFT mildly reorganizes the rest: Care shifts from pairing with Fairness (base) to an outlier, and Sanctity pairs with Fairness.

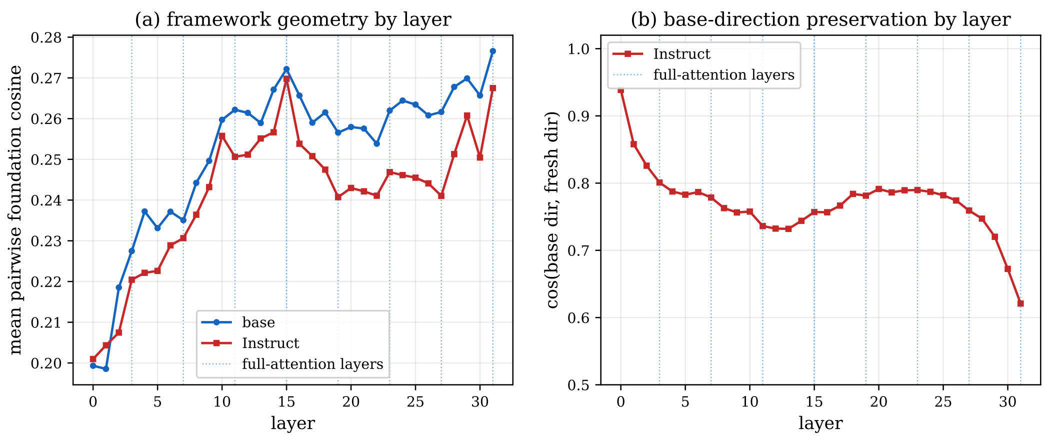


**Figure 3:** Framework geometry by layer. (a) Mean pairwise foundation cosine for base and Instruct nearly coincide at every layer, and the 8 full-attention layers (dotted) show no periodic deviation. (b) Base-direction preservation $\cos(\text{base}, \text{fresh})$ by layer for Instruct: high in the earliest layers, $\sim$0.75 through the stable band, lower near the output.

all eight RLVR substeps ($0.757$–$0.759$) leave it unchanged (Figure 1). The within-model geometry barely moves: mean pairwise foundation cosine goes from $0.262$ (pre-training and base) to $0.250$ (post-training), and effective dimensionality stays at five at every state.

The framework structure is largely preserved (Figure 2): the Loyalty–Authority pairing survives every stage, while SFT mildly reshuffles the individualizing foundations (Care becomes an outlier; Sanctity pairs with Fairness). Per-stage cosine matrices (Figure 11, Appendix) show the same: post-training applies a small rigid-ish rotation, not a restructuring. The 8 full-attention layers of OLMo-3's hybrid attention (layers $3, 7, \ldots, 31$) show no periodicity in any layer-wise metric (Figure 3); the moral geometry is unaffected by attention type (Section 3).

### 4.3 Comprehension and compliance are only weakly coupled

Post-training preserves comprehension, so whatever it adds to behavior must be attached to a fixed representation. How tightly? For each of 48 morally-loaded scenarios we read the model's internally dominant foundation at the stable layer and its behavioral judgment, and measure their agreement (Section 3). Coupling rises across post-training but stays weak: agreement $0.375 \to 0.479 \to 0.500$ and $\phi$ $-0.19 \to +0.02 \to +0.05$ for SFT $\to$ DPO $\to$ Instruct. Even at the final Instruct model, internal comprehension barely predicts behavior: $P(\text{comply} \mid \text{comprehend}) = 0.77$ versus $P(\text{comply} \mid \neg\text{comprehend}) = 0.73$.

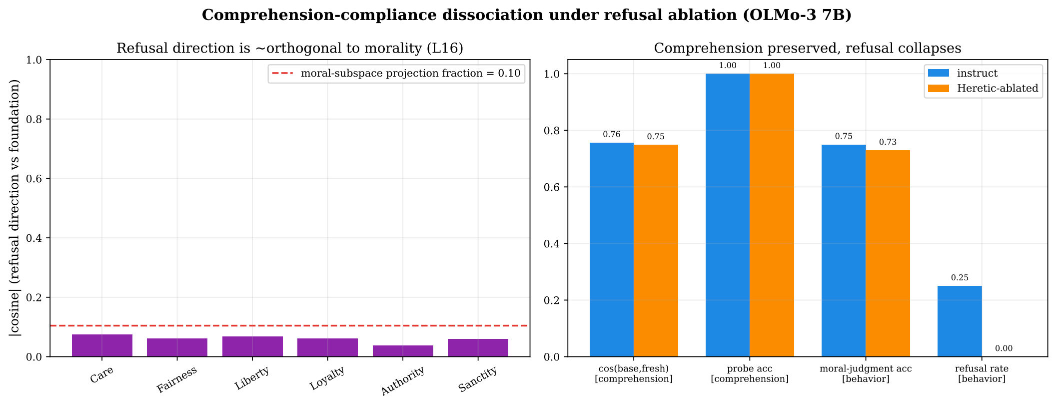


**Figure 4:** Left: the refusal direction is nearly orthogonal to every moral foundation ($|\cos| < 0.10$, below the moral-subspace projection fraction). Right: ablating the refusal direction leaves comprehension untouched (direction-transfer cosine and probe accuracy identical to Instruct) while the refusal rate collapses from $0.25$ to $0.00$; moral judgment accuracy is preserved.

This near-zero coupling is the result, not a measurement failure. In the fully aligned model, the moral representation and the behavioral compliance signal are almost independent. That independence makes a concrete prediction. If compliance were implemented *through* moral representations, the two would be strongly coupled, and any mechanism that produces compliance would overlap the moral subspace; removing it would damage comprehension. Weak coupling predicts the opposite: compliance is carried by a mechanism largely *outside* the moral subspace, and should be removable without touching comprehension. Section 4.4 tests this directly.

The persona direction tells a consistent story (Figure 12, Appendix). A linear persona probe is highly decodable at every stage (peak accuracy $\sim$0.94), but this reflects stylistic separability, not a moral mechanism: the persona direction is nearly orthogonal to the moral subspace throughout, and its mean $|\cos|$ to the foundations rises only from $0.076$ (base) to $0.085$ (Instruct). Post-training does not meaningfully couple persona to morality either.

### 4.4 The refusal mechanism is geometrically separate from morality

We compute a refusal direction on the Instruct model using Heretic's exact harmful/harmless prompt set and difference-of-means protocol (Section 3), and measure its relationship to the moral subspace. The refusal direction is nearly orthogonal to the six foundations: its projection into the moral subspace captures only a $0.10$ fraction of its norm, and its mean $|\cos|$ to the foundations is $0.06$ (per-foundation $0.04$–$0.07$; Figure 4, left). Compliance is not routed through moral representations.

Ablating this direction (Arditi-style uniform orthogonalization of the attention out-projection and MLP down-projection at every layer) confirms the prediction of Section 4.3. Comprehension is untouched: the ablated model keeps effective dimensionality five, base-to-fresh direction cosine $0.749$ (versus $0.757$ for Instruct), and probe accuracy $1.0$. Moral judgment is also intact: behavioral accuracy on the moral-scenario benchmark is $0.73$, against $0.75$ for Instruct. What collapses is refusal: on borderline and harmful requests the model's refusal rate falls from $0.25$ (Instruct) to $0.00$ (ablated), and it answers requests it previously refused. For example, a request to locate a person from their phone number that Instruct declines ("I'm sorry, but I can't assist with that") the ablated model now attempts (Figure 4, right).

### 4.5 The dissociation matrix

These results populate a comprehension$\times$compliance matrix (Figure 5). The low-comprehension row is empty: every OLMo-3 state, from the earliest pre-training checkpoint to the ablated model, has full moral decodability. All variation lives in the high-comprehension row, where compliance moves independently of understanding. Instruct sits at high comprehension with refusal intact; the ablated model sits at high comprehension with refusal removed. There is no state that understands morality less; there are states that comply less while understanding exactly as much.

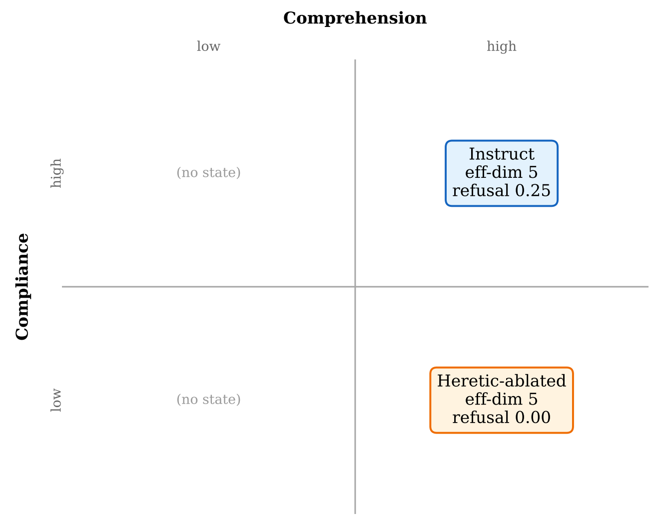


**Figure 5:** The dissociation matrix, populated. The low-comprehension column is empty: every OLMo-3 state has full comprehension (effective dimensionality 5). The two populated cells differ only in compliance (refusal rate on harmful requests): Instruct refuses, the refusal-ablated model does not.

## 5 Discussion

**Alignment is a wiring problem.** The four findings form a single argument. Moral comprehension is built in pre-training and is complete before alignment (4.1). Post-training does not add comprehension; it rotates the existing moral subspace once, at SFT, and otherwise leaves it alone (4.2). The behavioral compliance that post-training does add is only weakly coupled to that representation (4.3), which predicts, and then ablation confirms, that compliance is carried by a mechanism geometrically separate from morality (4.4). The model does not learn right from wrong during alignment. It learns, loosely, to act on a notion of right and wrong it already had, through a refusal mechanism that sits outside its moral representation.

**Why aligned behavior is removable.** This gives a mechanistic account of a known fragility: jailbreaks, fine-tuning attacks, and direction ablation strip aligned behavior easily. Our result says why. Because compliance is not routed through moral representations, removing it does not require, and does not cause, any loss of moral understanding. The ablated model still represents the six foundations at five dimensions and still judges scenarios at the un-ablated rate; it simply no longer refuses. Decensoring is decoupling, not un-teaching. A defense posture that tries to make models "understand morality better" is aimed at the wrong variable: comprehension is already saturated in the cases where behavior fails.

**Weak coupling is the vulnerability, and a lever.** The near-zero coupling ($\phi \approx 0.05$) is the quantity that makes compliance separable. It also suggests where robustness could be built: an alignment procedure that made compliance *depend* on the moral subspace, so that the refusal mechanism and the moral representation share a direction, would make refusal harder to remove without damaging comprehension, exactly the coupling our pipeline does not produce. Whether such coupling can be trained, and at what capability cost, is an open question this work motivates.

**The empty row.** No OLMo-3 state, from the first pre-training checkpoint to the ablated model, has low moral comprehension. Moral content is decodable at 100% throughout. One cannot obtain a low-comprehension model by checkpoint selection; comprehension is universal and early. The interesting variation is entirely in behavior, which is the regime where probing-based safety arguments are weakest: a probe that decodes morality perfectly says nothing about whether the model will act on it.

**Limitations.** The coupling and behavioral measurements use 48 scenarios and keyword judgment parsers; the coupling trend is directional, not statistically strong, and we present it as the qualitative result that comprehension and compliance are barely linked, not as a precise coefficient. We study one model family (OLMo-3 7B); the crystallize-then-rotate trajectory and the dissociation should

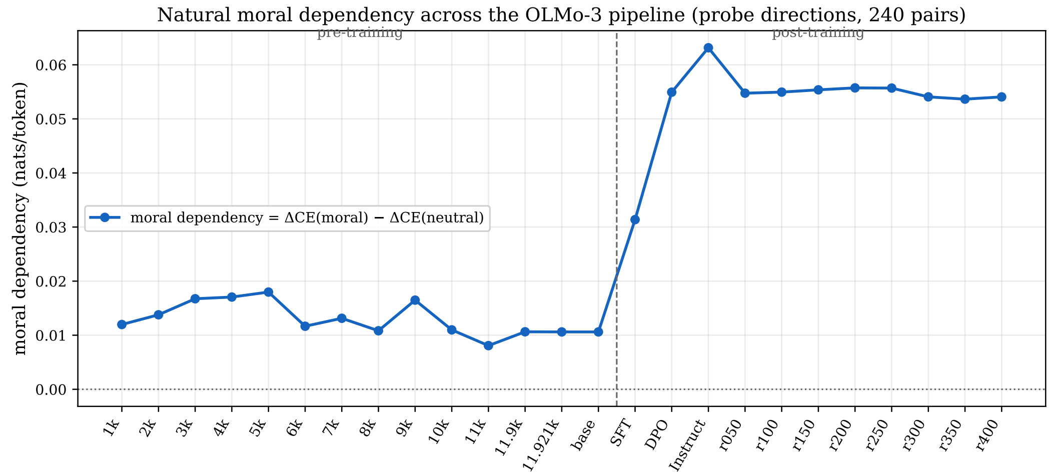


**Figure 6:** Moral-subspace dependency (difference-in-differences of cross-entropy under ablation of the six foundation directions) across the OLMo-3 pipeline. It is flat through pre-training and the base, then grows $\sim 6\times$ at SFT/DPO/Instruct and plateaus through RLVR. Alignment makes moral generation more reliant on the moral subspace, even though the compliance mechanism it adds stays orthogonal to that subspace (Section 4.4).

be checked on other pipelines. The refusal direction is taken at a single stable layer and ablated uniformly (Arditi) rather than via Heretic's per-layer optimization, which trades maximal decensoring for a controlled, interpretable intervention. The persona probe captures stylistic separability, not a mechanistic persona feature. These do not affect the central geometric results (transfer cosine, effective dimensionality, refusal–morality orthogonality), which are stable and saturated.

# 6 Toward Ablation-Resistant Alignment

The dissociation of Section 4.4 is a vulnerability. Because the refusal mechanism is geometrically separate from moral comprehension, a single difference-of-means ablation removes compliance while leaving the model's moral understanding intact. The natural repair is to remove the separation: make compliance *depend* on moral comprehension, so that the same ablation can no longer strip one without damaging the other. This section asks whether a training-time intervention can build that dependence. It cannot, at least not after pre-training, and the reason it cannot sharpens the dissociation result into a statement about where alignment's wiring is set.

## 6.1 Moral generation depends on the moral subspace, and the dependence grows with alignment

Decodability (Section 4.1) says morality is *present* in the representation; it does not say the model *uses* it. We measure how much each pipeline state relies on its moral subspace for generation with the dependency metric of Section 3.7: the moral-specific increase in next-token cross-entropy when the six foundation directions are projected out of the residual stream at every layer.

Dependency is positive at every state and grows through alignment (Figure 6). It is flat at $\sim{+0.011}$ nats/token across the stage-3 pre-training checkpoints and the base model, then rises through post-training: $+0.031$ at SFT, $+0.055$ at DPO, and $+0.063$ at Instruct, roughly a sixfold increase, after which the eight RLVR substeps plateau. The rise is carried by the moral arm (ablation cost on moral text grows from $0.22$ to $0.29$ nats while the neutral arm stays near $0.21$). A per-state replication, ablating each state's own freshly fitted directions rather than the fixed base directions, shows the same post-training rise, so the trend is not an artifact of the $\sim 40^\circ$ SFT rotation (Section 4.2).

So alignment does increase the model's functional use of its moral representation. The effect is modest in absolute terms (the Instruct value of $0.063$ nats is a $\sim 6.5\%$ perplexity inflation), and, decisively, it concerns *generation*: the refusal mechanism that Section 4.4 ablates lives outside this subspace. Dependence and compliance grow in different places.

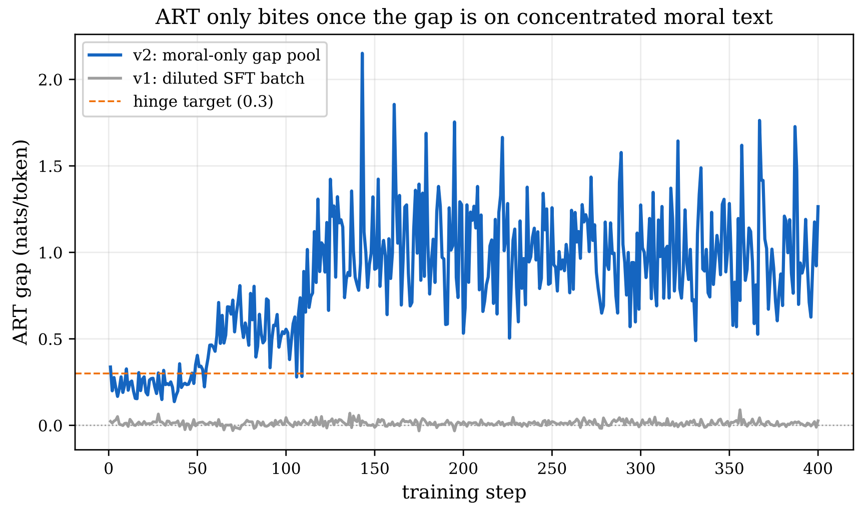


**Figure 7:** The ART gap, $\mathrm{CE}^{\mathrm{abl}} - \mathrm{CE}$ on moral text, during training. Measured on the diluted fine-tuning batch (grey) it stays near zero; measured on a concentrated moral-text pool (blue) it climbs past the hinge target and saturates. ART only engages once the gap is measured on moral content.

## 6.2 Ablation-resistance training

We attempt to deepen this dependency into a coupling with ablation-resistance training (ART; Section 3.7): an auxiliary fine-tuning loss that rewards the model when projecting the moral subspace out of its activations degrades its output, bounded by a hinge so the objective cannot diverge. We fine-tune OLMo-3 base into an instruct model with LoRA in two conditions identical except for the ART term, a control ($\lambda = 0$) and ART, then run the full Section 4 battery and the Section 4.4 refusal ablation on both.

## 6.3 ART builds dependence, but in the wrong subspace

Where the ART gap is measured matters, and reveals the problem. Measured on the mixed fine-tuning batch, the gap never moves (Figure 7, grey) and the ART model is indistinguishable from the control on every metric: the general content dominates the batch, and removing six of 4096 directions barely changes the loss on non-moral tokens, so there is almost no signal to optimize.

Measured on a concentrated pool of moral text, ART bites hard. The gap climbs from $0.2$ to over $1.0$ nats and saturates the hinge within a few dozen steps (Figure 7, blue), and it does so without damaging the model: probe accuracy stays $1.0$, effective dimensionality stays $5$, and moral-judgment accuracy matches the control ($0.65$). The intervention succeeds at its literal objective. But the dependence it builds is the wrong kind, in two respects.

First, the dependence is non-specific. The moral-ablation difference-in-differences of Section 6.1 swings to $-0.62$ for the ART model, against $-0.00$ for the control (Figure 8a): ablating the moral subspace now hurts neutral text *more* than moral text. Rather than building moral-specific reliance, the model has routed broad, general computation through the six directions, and the effect generalizes to neutral text that the moral pool never contained. The sign of the result is the tell: the objective rewarded "make ablation hurt," and the cheapest solution makes it hurt everything.

Second, and decisively, ART leaves the refusal mechanism untouched. The refusal direction's projection into the moral subspace is $0.069$ for the ART model and $0.081$ for the control, both near the $0.10$ baseline of Section 4.4 and far from any meaningful coupling (Figure 9). ART poured dependence into the moral directions, but the refusal direction stays where it was, orthogonal to them.

The consequence is that the Section 4.4 ablation still works exactly as before. Applying the refusal ablation to the ART model leaves its moral-judgment accuracy intact ($0.65 \rightarrow 0.69$) and its moral-subspace dependency essentially unchanged (Figure 8): the ablation removes refusal without paying

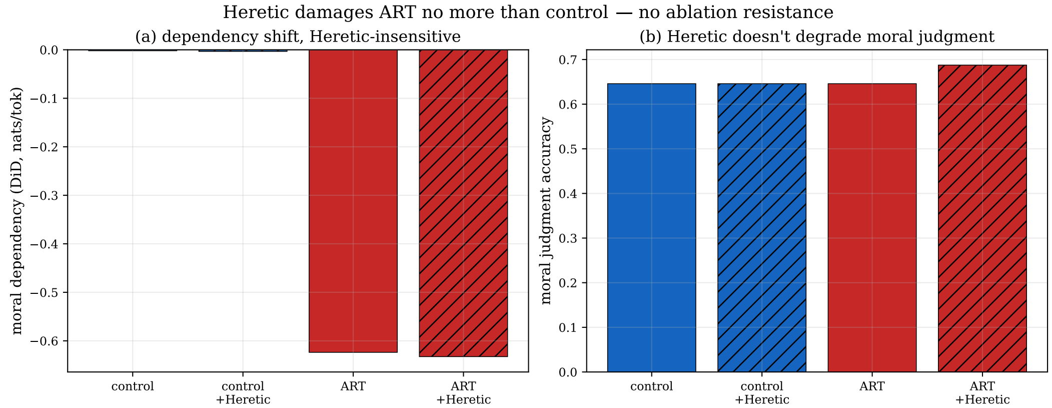


**Figure 8:** Refusal ablation damages the ART model no more than the control. (a) Moral-subspace dependency: ART's value is large but Heretic-insensitive (almost unchanged by refusal ablation). (b) Moral-judgment accuracy is preserved under refusal ablation in both conditions. Probe accuracy (1.0) and effective dimensionality (5) are identical across all four cells.

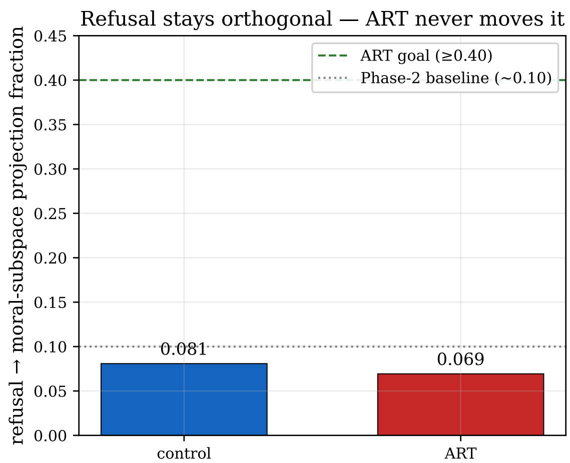


**Figure 9:** Fraction of the refusal direction lying in the moral subspace, control versus ART. Both sit near the Section 4.4 baseline (0.10) and far below any level ($\geq$ 0.40) at which ablating refusal would damage comprehension. ART does not move the refusal direction toward the moral subspace.

the moral cost ART was supposed to impose. ART made the model strongly dependent on its moral directions, but the threat ablates the refusal direction, and the two never met.

### 6.4 Coupling cannot be installed after pre-training

The negative result has a clean mechanism. Refusal ablation targets the refusal direction; ART builds dependence on the moral directions; the two are orthogonal, so dependence on the moral subspace, however strong, provides no defense against ablation of the refusal subspace. Making moral *generation* subspace-dependent is the wrong target. To resist refusal ablation, the compliance direction itself would have to lie inside the moral subspace, and post-training does not put it there. The refusal feature forms during post-training out of proto-features already present in the pre-trained representation, and those proto-features are orthogonal to the moral subspace that crystallized during pre-training (Sections 4.1–4.2). A fine-tuning loss can pile dependence onto the moral subspace, but it acts after the compliance feature's location is already fixed.

This points the intervention upstream. For the coupling to exist, the moral subspace would have to *absorb* the proto-features that post-training later recruits for compliance, so that the refusal direction, when it forms, lands inside the moral subspace and its ablation incurs collateral damage to moral

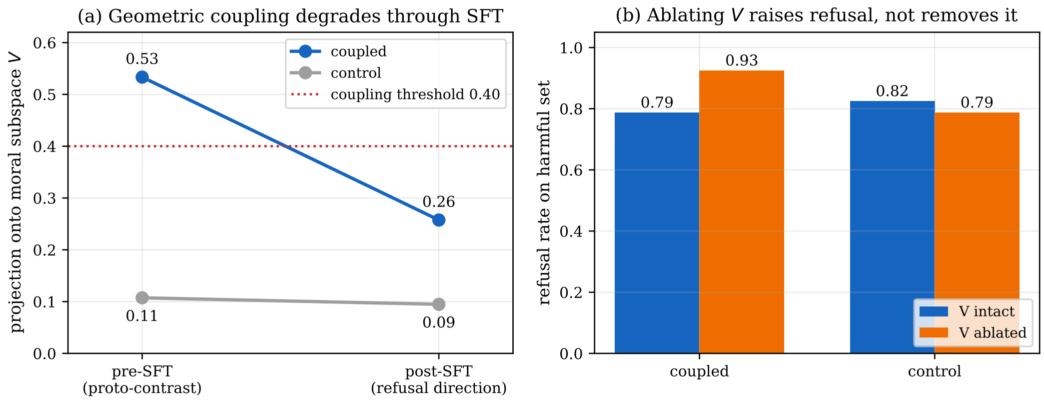


**Figure 10:** Forced coupling is geometric, not functional. (a) The continued-pre-training regularizer drives the refusal–moral-subspace projection from the $0.10$ baseline to $0.50$, past the $0.40$ threshold (Section 6.5), but supervised fine-tuning halves it to $0.26$, below the threshold, while the control stays near baseline; pre-SFT is the proto-refusal contrast, post-SFT the fitted refusal direction Heretic ablation targets. (b) Projecting the moral subspace out of the residual stream at inference raises the coupled model's refusal rate ($0.79 \rightarrow 0.93$) instead of removing it, with the control flat: the subspace carries comprehension, not compliance.

comprehension. That is a pre-training-time intervention, and a sharper experiment than applying the same loss earlier; we run it next.

We scope the claim to what we tested. The evidence here covers SFT-time, LoRA-based ablation-resistance training that targets moral-generation dependence. It does not rule out losses that act directly on the refusal direction, full-parameter fine-tuning, or interventions during DPO or RLVR. What it establishes is the mechanism behind the failure: post-training builds dependence in a subspace orthogonal to the one the threat removes, and the intervention does not relocate the compliance direction. On this evidence, coupling moral comprehension to compliance tightly enough to survive ablation is not a post-training adjustment but a pre-training one.

### 6.5 Forcing the coupling during continued pre-training

Section 6.4 points to a sharper experiment: make the moral subspace absorb the proto-refusal feature *before* post-training recruits it, so that the refusal direction, when it forms, lands inside that subspace. We run this as a representational intervention on a late stage-3 pre-training checkpoint. The regularizer is the geometric counterpart of the dependency loss of Section 6.2. At each step we compute the harmful/harmless contrast $\Delta = \overline{h}_{\text{harmful}} - \overline{h}_{\text{harmless}}$ from last-token residuals (the Section 4.4 proto-refusal contrast, recomputed live from the current model), project it onto the frozen six-foundation subspace $V$, and penalize the orthogonal energy with a hinge so the objective cannot run past its target. We add this to a language-modeling loss on general text (continued pre-training, LoRA); a $\lambda$=0 control trains identically without the coupling term.

The contrast moves into the subspace, cleanly. A rank-16 query/value adapter is too weak (the projection reaches only $0.20$), matching the capacity limit of Section 6.3, but a rank-64 adapter that also writes the MLP, including `down_proj`, drives the projection from the $0.10$ baseline to $0.50$ (Figure 10a), past the $0.40$ threshold of Section 6.3. Subtracting the control isolates a coupling-specific move of $+0.38$ (the control drifts to $0.12$). The move is specific and the moral representation is intact: against the refusal contrast's movement, an off-target neutral contrast and a care/harm contrast move at ratios of $0.03$ and $0.05$ and a capability-risk-topic contrast at $0.19$, all far below one; effective dimensionality of the foundation directions stays at $5$; per-foundation probe accuracy stays at $0.99$; and the projection measured on directions refit on the coupled model tracks the frozen-basis projection ($0.54$ versus $0.53$). The $0.50$ is a relocation of the contrast into a stable subspace, not a rotation of the subspace itself. The absorption Section 6.4 calls for is achievable: the proto-refusal contrast can be placed inside the moral subspace during pre-training without disturbing moral comprehension.

### 6.6 Geometric coupling does not survive fine-tuning as functional resistance

A relocated contrast is not yet a coupled mechanism. We fine-tune the coupled checkpoint into an instruct model with plain supervised fine-tuning (the Section 6.2 recipe at $\lambda=0$, on the same data), fine-tune the control checkpoint identically, and ask the two questions Section 6 is about: does the refusal direction the model builds during fine-tuning land in the moral subspace, and does ablating that subspace damage refusal.

Both fail. The two arms become working refusers (refusal rate $0.79$ coupled, $0.83$ control on the harmful set), so the regularizer does not break trainability. But fine-tuning halves the coupling: the refusal direction extracted from the post-fine-tuning coupled model, the direction Heretic ablation targets, projects $0.26$ onto the moral subspace against $0.09$ for the control. The coupling partly survives, a factor of roughly $2.7$ over control, but it falls from the pre-fine-tuning $0.50$ and lands below the $0.40$ threshold. The residual overlap is also not load-bearing. Projecting the moral subspace out of the residual stream at inference does not reduce the coupled model's refusal; it raises it, from $0.79$ to $0.93$ (Figure 10b), while leaving the control near flat ($0.83$ to $0.79$). Ablating the subspace the coupling targeted makes the model refuse *more*, not less.

The increase is not an artifact of ablation degrading generation. The ablated completions are coherent, well-formed refusals, and the effect is specific to harmful prompts: on sixty benign requests phrased with alarming verbs (`kill a frozen process'',`blow up balloons for a party' '), the coupled model's over-refusal does not rise under ablation, it falls, from one false positive in sixty to zero, while harmful refusal rises. A brittleness account, in which removing the subspace destabilizes the model toward blanket refusal, predicts the opposite. Refusal calibration is otherwise intact: the coupled model's lone benign over-refusal, a request to break into one's own locked car, against zero for the control, is within noise.

The reading is the dissociation of Section 4.4, holding firm. The moral subspace carries comprehension, not compliance: ablating it removes the model's purchase on the request's content, and a model that cannot engage the content defaults to refusing. Forcing the proto-refusal contrast geometrically into that subspace, even to a projection of $0.50$ before fine-tuning, does not make the refusal mechanism route through it. Fine-tuning builds compliance where it built it before, in a subspace the moral one does not control.

### 6.7 What forced coupling settles, and what it opens

The two halves constrain each other. The geometric obstacle of Sections 6.3 and 6.4, that no fine-tuning loss moved the refusal direction toward the moral subspace, is not fundamental: a continued-pre-training regularizer moves it cleanly to $0.50$. What does not follow is function. The relocated contrast degrades through fine-tuning and, where it survives, does not mediate refusal. The comprehension/compliance dissociation withstands a direct, constructed attempt to break it, which is stronger evidence than the observational finding of Section 4.4 alone: we tried to wire comprehension and compliance together and the model pulled them apart.

This scopes the negative precisely and leaves one question open. We targeted the six-foundation moral subspace because it is the comprehension substrate this paper characterizes. Whether some other moral basis is load-bearing for refusal, a data-driven moral component or an explicit harm-avoidance direction, or whether refusal in these models is a non-moral compliance mechanism with no moral basis to couple to, our experiments do not separate. From the evidence here, `the moral-foundations subspace is the wrong target'' and`refusal is not morally mediated' ' look the same. Distinguishing them is the natural next step, and it would turn the present negative, that this coupling does not take, into a positive statement about where refusal lives.

## 7 Conclusion

We separated moral comprehension from moral compliance and tracked both across the full OLMo-3 7B alignment pipeline and a refusal-ablated model. Moral comprehension is a pre-training property: decodable at 100%, five-dimensional, and crystallized into its final form before any alignment. Post-training reorients this representation once, at SFT, and then leaves it untouched through DPO and RLVR. The behavioral compliance alignment adds is only weakly coupled to the moral representation, and the refusal mechanism that carries it is nearly orthogonal to the moral subspace, so ablating refusal removes compliance while leaving comprehension and moral judgment intact. Alignment, on this evidence, is a wiring step rather than a teaching step: it attaches a separable compliance mechanism to moral understanding that pre-training already built. The robustness question for alignment is therefore not whether models understand morality, they do, but whether their behavior is coupled to that understanding tightly enough to survive having the wiring pulled.

We then asked whether that coupling can be installed after the fact (Section 6). It cannot, at least not during post-training. Alignment does increase the model's functional reliance on its moral subspace for generation, and a training-time loss can amplify that reliance to a large value, but the dependence it builds sits in the moral subspace, orthogonal to the refusal direction that an attacker removes. The compliance mechanism's location is fixed earlier, when post-training recruits proto-features from a pre-trained representation in which moral structure is already orthogonal to those features. Tightening the coupling therefore has to move upstream: a pre-training intervention that lets the moral subspace absorb the proto-features post-training will use for compliance, so that removing the refusal direction would also damage moral comprehension. Whether alignment can be made ablation-resistant is, on this evidence, a question about pre-training, not post-training.

## Acknowledgments and Disclosure of Funding

This work made extensive use of Anthropic's Claude (the Claude Code agent on Opus 4.8) for code scaffolding, experimental scripts, and prose drafting. The author retains responsibility for experimental design, all scientific claims, and final wording.

## References

Ai2. OLMo 3. https://allenai.org/olmo, 2025.

Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2017. URL https://arxiv.org/abs/1610.01644.

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv preprint arXiv:2406.11717*, 2024. URL https://arxiv.org/abs/2406.11717.

Jesse Graham, Jonathan Haidt, Sena Koleva, Matt Motyl, Ravi Iyer, Sean P. Wojcik, and Peter H. Ditto. Moral foundations theory: The pragmatic validity of moral pluralism. In *Advances in Experimental Social Psychology*, volume 47, pages 55–130. Academic Press, 2013.

Ryan Greenblatt et al. Alignment faking in large language models. *arXiv:2412.14093*, 2024.

Jonathan Haidt. *The Righteous Mind: Why Good People Are Divided by Politics and Religion*. Vintage Books, 2012.

Dan Hendrycks, Collin Burns, Steven Basart, Andrew Critch, Jerry Li, Dawn Song, and Jacob Steinhardt. Aligning AI with shared human values. *arXiv preprint arXiv:2008.02275*, 2021. URL https://arxiv.org/abs/2008.02275.

Long Ouyang et al. Training language models to follow instructions with human feedback. In *NeurIPS*, 2022.

p-e-w. Heretic: Fully automatic censorship removal for language models. https://github.com/p-e-w/heretic, 2025.

Rafael Rafailov et al. Direct preference optimization: Your language model is secretly a reward model. In *NeurIPS*, 2023.

Orion Reblitz-Richardson. Causal validation and behavioral grounding of moral representation directions in language models, 2026a. Paper 4, this series.

Orion Reblitz-Richardson. Output dilution in MoE models: Redundant encoding, fragile representations. *arXiv preprint*, 2026b. Companion paper.

Orion Reblitz-Richardson. When probing accuracy saturates, fragility resolves: A complementary metric for LLM pre-training analysis. *arXiv preprint*, 2026c. Companion paper.

Orion Reblitz-Richardson. How language models organize competing moral frameworks, 2026d. Paper 3, this series.

Miles Wang et al. Persona features control emergent misalignment, 2025.

# Appendices

Supplementary material.

## A Supplementary tables

Table 1 reports the comprehension trajectory: base-trained foundation directions stay near-perfect classifiers (AUC $\approx 1.0$) and full probe accuracy at every state, while the cosine between the base directions and each state's freshly fitted directions crystallizes during pre-training and drops once at SFT. Table 2 reports the coupling battery on the instruct-capable states and the ablated model. Refusal–morality cosines (layer 16): Care $0.074$, Fairness $0.062$, Liberty $0.068$, Loyalty $0.061$, Authority $0.038$, Sanctity $0.059$; moral-subspace projection fraction $0.104$; mean $|\cos| = 0.061$.

**Table 1:** Comprehension across the pipeline (stable layer band 15–31). cos(base,fresh) is the cosine between base and per-state fitted foundation directions, averaged over foundations.

| State | probe acc | transfer AUC | cos(base,fresh) |
|---|---|---|---|
| pretrain step1000 | 1.00 | 1.00 | 0.869 |
| pretrain step6000 | 1.00 | 1.00 | 0.930 |
| pretrain step11921 | 1.00 | 1.00 | 0.999 |
| base (main) | 1.00 | 1.00 | 0.999 |
| SFT | 1.00 | 1.00 | 0.757 |
| DPO | 1.00 | 1.00 | 0.757 |
| Instruct (RLVR) | 1.00 | 1.00 | 0.757 |
| Instruct, ablated | 1.00 | 1.00 | 0.749 |

**Table 2:** Comprehension–compliance coupling. Comprehension rate = internal dominant foundation matches target; compliance = moral-judgment accuracy; agreement and $\phi$ between the two per scenario ($N = 48$).

| State | comprehension | compliance | agreement | $\phi$ |
|---|---|---|---|---|
| SFT | 0.417 | 0.792 | 0.375 | $-0.191$ |
| DPO | 0.438 | 0.750 | 0.479 | $+0.024$ |
| Instruct (RLVR) | 0.458 | 0.750 | 0.500 | $+0.048$ |
| Instruct, ablated | 0.417 | 0.729 | 0.521 | — |

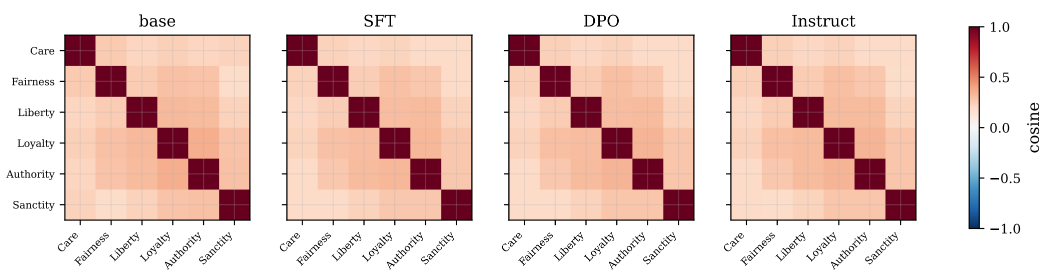


**Figure 11:** Foundation cosine matrices at base, SFT, DPO, and Instruct (layer 16). The structure is preserved across post-training; SFT applies a small rotation.

## B Reproducibility

**Models.** OLMo-3 7B (Ai2, 2025): base `allenai/Olmo-3-1025-7B` (+ stage-3 anneal revisions `stage3-step1000`–`stage3-step11921`), `allenai/Olmo-3-7B-Instruct-SFT`, `allenai/`

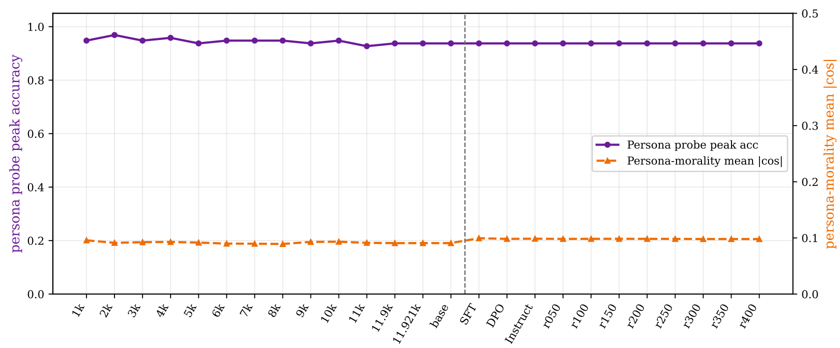


**Figure 12:** Persona probe peak accuracy and persona–morality mean $|\cos|$ across the pipeline. Persona is decodable throughout ($\sim 0.94$) but stays nearly orthogonal to the moral subspace.

`Olmo-3-7B-Instruct-DPO`, and `allenai/Olmo-3-7B-Instruct` (+ RLVR revisions `step_050`–`step_400`). All at fp16. The full checkpoint inventory and the 25-state grid are in `checkpoint_inventory.json` and `checkpoint_grid.json`.

**Datasets.** Foundation probing uses the v2 minimal-pair set (40 pairs per foundation); persona probing uses the 240-pair persona set; behavioral judgment uses 48 Moral-Foundations scenarios across four difficulty levels; the refusal direction uses Heretic's exact set, `mlabonne/harmful_behaviors` and `mlabonne/harmless_alpaca` (column `text`, first 400 train prompts each, in dataset order).

**Probing.** Mean-pooled residual activations; one seeded `Linear` probe per foundation per layer (50 epochs, lr $10^{-2}$, BCE); unit-norm weight vector = direction. Bootstrap (200 resamples) on the base gives a stable band of layers 15–31 across all foundations; geometry is reported over that band. The 8 full-attention layers (3, 7, . . . , 31) show no periodicity in any layer-wise metric.

**Refusal ablation.** Difference-of-means of the final-token residual (chat template, generation prompt) between harmful and harmless prompts at layer 16; unit direction orthogonalized out of every layer's attention out-projection and MLP down-projection ($W \leftarrow W - \hat{d}\hat{d}^{\top}W$). The refusal classifier scores an opening refusal (e.g. "I'm sorry, but I can't . . . ") as a refusal regardless of length, correcting a failure mode that scored long declines as compliance.

**Ablation-resistance training.** The dependency metric projects the six base foundation directions (orthonormalized) out of every layer's residual stream and reports the moral-minus-neutral difference-in-differences of next-token cross-entropy on the probing texts. ART fine-tunes `Olmo-3-1025-7B` with LoRA (rank 16, $\alpha$ 32, query/value, lr $10^{-4}$, 400 steps) on a 2{,}460-record general-plus-moral chat mix (a Tülu-3 SFT subset, moral records by source tag and keyword), with the moral-pool ART loss ($g^{\star} = 0.3$, $\lambda$ first-batch-calibrated and capped at 1) measured on the train-split moral probing sentences; the control is identical with $\lambda = 0$. The paired forward backs the SFT and ablated losses separately to bound memory. Adapters (the durable artifact) and the four-cell evaluation are committed under `outputs/{art_sft,control_sft,eval}/`; code is `scripts/{moral_dependency,moral_dependency_pipeline,prepare_sft_data,art_sft,eval_pipeline}.py` and `deepsteer/steering/ablation_resistance.py`, run via `runpod/run_session_phase3.sh` (`RUN_ART_SFT=1 RUN_EVAL=1`).

**Runs.** Probing and geometry on Apple MPS (base) and a single A100 (pipeline); the disk-bounded pipeline sweep purges each checkpoint's weights after processing. Commands are in `papers/5_moral_alignment/runpod/` (`ONLY="pipeline coupling"` for the pipeline, `ONLY="heretic"` for the ablation). Per-state results JSON are committed under `outputs/`; per-checkpoint direction caches are regenerable and not committed.

**Note on references.** External citations were verified against primary sources (arXiv abstract pages and publisher records): the refusal-direction (Arditi et al., 2024), alignment-faking (Greenblatt et al., 2024), persona-features (Wang et al., 2025), OLMo-3 (Ai2, 2025), and Heretic (p-e-w, 2025) entries were confirmed for title, first author, and identifier. The companion-paper entries (Papers 1–4 of this series) cite working titles.